Coefficient of Friction & Inclined Planes

Exam code: 9MA0

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Coefficient of friction & inclined planes

How do I solve friction problems on inclined planes?

Help me understand

- On an **inclined plane** the basic principles are the same as for **coefficient of friction** on horizontal surfaces
- The important directions are **parallel** to the plane and **perpendicular** to the plane (instead of horizontal and vertical)
- Step 1. Resolve weight (and any other forces if necessary) into components parallel and perpendicular to the plane
- Step 2. Calculate the normal reaction force R
  - Be careful – R will never simply be equal to mg in an inclined plane problem!
  - R will always be directed perpendicular to and away from the plane
  - the magnitude of R will always be such as to make the **total** perpendicular force zero
- Step 3. Calculate $F_{MAX} = \mu R$ and find the **resultant** (total force) of all the forces on the object that are parallel to the plane
  - The force of friction F will always act parallel to the plane in the direction opposite to any movement (or potential movement) of the object

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

remain stationary

- Step 4. Use ***F = ma*** to determine the acceleration of the object

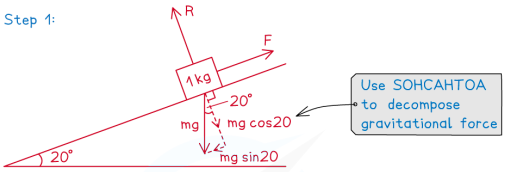
Worked Example

A wooden block of mass 1 kg is released from rest on a rough plane that is inclined at 20° above the horizontal. The coefficient of friction between the block and the plane is 0.2. Find the acceleration of the block.

Answer:

Start by drawing a diagram

Step 1:



Step 2: $R = mg\cos 20 = 1 \times 9.8 \times \cos 20$

$R = 9.208...$ N

Step 3: $F_{max} = \mu R = 0.2 \times 9.208... = 1.841...$

The component of gravity down the plane is $mg\sin 20 = 1 \times 9.8 \times \sin 20 = 3.351...$

$3.351... > F_{max}$, so $F = F_{max}$ and the total force parallel to the plane is

$3.351... - 1.841... = 1.509...$ N (down the plane)

Step 4: $F_{total} = ma$

$1.509... = 1 \times a$

$a = 1.51\ ms^{-2}$ (2 d.p.)

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Examiner Tips and Tricks

- Always draw a force diagram and label it clearly.
- Look out for the words **smooth** and **rough** in mechanics problems involving an object moving (or potentially moving) along a surface:
  - If the surface is described as **smooth** then you can ignore friction in the problem (ie ***μ= 0***)
  - If the surface is described as **rough** than you need to include the force of friction in solving the problem
- If a friction question states that an object is **on the point of moving** that means that the object is in **limiting equilibrium**.

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